

COMMERCIAL SOLAR — FINANCIAL ANALYSIS

Returns, plant and tax depreciation

450 kWp
PLANT CAPACITY

APSPDCL · COMMERCIAL ROOFTOP

450 KWP GRID-TIED SOLAR PLANT

Net investment ₹1.74 Cr after input tax credit, generating 6,89,850 units in year one against an energy charge of ₹7.65 per unit.

HT CATEGORY I / II AT 11 KV

30-YEAR ANALYSIS

PAYBACK

3.7 yrs

Post-tax, undiscounted

PROJECT IRR

27.5%

Unlevered, 30 years

COST OF SOLAR ENERGY

₹3.15

per kWh, against ₹7.65 from the grid

NET PRESENT VALUE

₹3.20 Cr

Discounted at 10%

PLANT AND GENERATION

Plant capacity	450 kWp
Year 1 generation	6,89,850 kWh
Specific yield	1,533 kWh / kWp / year
Supply category	HT Category I / II at 11 kV (HT supply at 11 kV)
Energy charge offset	₹7.65 / kWh
Units self-consumed	90% at the energy charge
Units exported	10% at ₹2.75 / kWh (APPC)
Year 1 gross saving	₹49.39 L

INVESTMENT AND TAX BASIS

Price per watt-peak	₹40.50 / Wp (incl. GST)
Project price incl. GST	₹1.82 Cr
GST at 5% — recovered as ITC	₹8.68 L
Net capitalised cost	₹1.74 Cr
Depreciation block	40% WDV, Appendix I
Tax regime	s.115BAA — 25.168%
Annual maintenance	₹600 / kWp / year, year 1 free
30-year depreciation shield	₹43.68 L

ACCELERATED DEPRECIATION — FIRST 12 YEARS

YEAR	OPENING WDV (₹)	DEPRECIATION AT 40% (₹)	CLOSING WDV (₹)	TAX SHIELD AT 25.168% (₹)
1	1,73,57,143	69,42,857	1,04,14,286	17,47,378
2	1,04,14,285	41,65,714	62,48,571	10,48,427
3	62,48,572	24,99,429	37,49,143	6,29,056
4	37,49,143	14,99,657	22,49,486	3,77,434
5	22,49,485	8,99,794	13,49,691	2,26,460
6	13,49,692	5,39,877	8,09,815	1,35,876
7	8,09,815	3,23,926	4,85,889	81,526
8	4,85,889	1,94,356	2,91,533	48,916
9	2,91,533	1,16,613	1,74,920	29,349
10	1,74,920	69,968	1,04,952	17,610
11	1,04,952	41,981	62,971	10,566
12	62,971	25,188	37,783	6,339
Tax shield, years 1-12				43,58,937

COMMERCIAL SOLAR — FINANCIAL ANALYSIS

30-year post-tax cash flow

450 kWp

PLANT CAPACITY

How this table works. Electricity savings reduce an operating expense, so they increase taxable profit and are taxed at 25.168%. Depreciation reduces the same taxable profit. Both are applied to one figure, so the tax column is a net inflow in the early years when depreciation exceeds savings — shown in green. Net cash flow is therefore gross saving, less maintenance, less that tax. The shaded row is the year the investment is recovered.

YEAR	GENERATION (KWH)	GROSS SAVING (₹)	AMC (₹)	DEPRECIATION (₹)	TAX (₹)	NET CASH FLOW (₹)	CUMULATIVE (₹)
1	6,89,850	49,39,326	—	69,42,857	(5,04,249)	54,43,575	54,43,575
2	6,86,401	51,11,215	2,70,000	41,65,714	1,70,010	46,71,205	1,01,14,780
3	6,82,969	52,89,085	2,70,000	24,99,429	6,34,147	43,84,938	1,44,99,718
4	6,79,554	54,73,145	2,70,000	14,99,657	9,32,094	42,71,051	1,87,70,769
5	6,76,156	56,63,610	2,70,000	8,99,794	11,31,004	42,62,607	2,30,33,376
6	6,72,775	58,60,704	2,70,000	5,39,877	12,71,192	43,19,512	2,73,52,888
7	6,69,411	60,64,657	2,70,000	3,23,926	13,76,873	44,17,783	3,17,70,671
8	6,66,064	62,75,707	2,70,000	1,94,356	14,62,601	45,43,106	3,63,13,777
9	6,62,734	64,94,101	2,70,000	1,16,613	15,37,133	46,86,969	4,10,00,746
10	6,59,420	67,20,096	2,70,000	69,968	16,05,751	48,44,345	4,58,45,091
11	6,56,123	69,53,955	2,70,000	41,981	16,71,652	50,12,303	5,08,57,394
12	6,52,843	71,95,953	2,70,000	25,188	17,36,784	51,89,169	5,60,46,563
13	6,49,578	74,46,372	2,70,000	15,113	18,02,346	53,74,026	6,14,20,589
14	6,46,331	77,05,506	2,70,000	9,068	18,69,086	55,66,420	6,69,87,009
15	6,43,099	79,73,657	2,70,000	5,441	19,37,487	57,66,170	7,27,53,179
16	6,39,883	82,51,141	2,70,000	3,264	20,07,872	59,73,269	7,87,26,448
17	6,36,684	85,38,280	2,70,000	1,959	20,80,468	61,87,813	8,49,14,261
18	6,33,501	88,35,413	2,70,000	1,175	21,55,447	64,09,965	9,13,24,226
19	6,30,333	91,42,885	2,70,000	705	22,32,950	66,39,935	9,79,64,161
20	6,27,181	94,61,057	2,70,000	423	23,13,099	68,77,958	10,48,42,119
21	6,24,046	97,90,302	2,70,000	254	23,96,006	71,24,296	11,19,66,415
22	6,20,925	1,01,31,005	2,70,000	152	24,81,779	73,79,225	11,93,45,640
23	6,17,821	1,04,83,564	2,70,000	91	25,70,527	76,43,037	12,69,88,677
24	6,14,732	1,08,48,392	2,70,000	55	26,62,356	79,16,036	13,49,04,713
25	6,11,658	1,12,25,916	2,70,000	33	27,57,377	81,98,539	14,31,03,252
26	6,08,600	1,16,16,577	2,70,000	20	28,55,702	84,90,876	15,15,94,128
27	6,05,557	1,20,20,834	2,70,000	12	29,57,447	87,93,387	16,03,87,515
28	6,02,529	1,24,39,159	2,70,000	7	30,62,732	91,06,427	16,94,93,942
29	5,99,516	1,28,72,042	2,70,000	4	31,71,681	94,30,361	17,89,24,303
30	5,96,519	1,33,19,989	2,70,000	3	32,84,421	97,65,569	18,86,89,872
Total	1,92,62,793	25,41,43,645	78,30,000	1,73,57,139	5,76,23,775	18,86,89,872	18,86,89,872

RETURN METRICS

Simple payback	3.7 yrs
Discounted payback	4.9 yrs
Project IRR, unlevered	27.5%
Net present value at 10%	₹3.20 Cr

OVER THE FULL TERM

Levelised cost of energy	₹3.15 / kWh
Cumulative return, 30 years	987%
Total net cash flow, after tax	₹18.87 Cr
30-year generation	1,92,62,793 kWh

COST OF A UNIT — SOLAR AGAINST THE GRID

Solar, levelised over 30 years		₹3.15
Grid energy charge today		₹7.65

Every unit the plant generates costs ₹3.15 across its life against ₹7.65 bought from APSPDCL today — and the grid figure reaches ₹23.86 by year 30 at 4% escalation, while the solar figure is fixed at the day of purchase. Over 30 years the plant generates 1,92,62,793 units, avoiding 15,795.5 tonnes of CO₂.

ASSUMPTIONS

- Tariff:** HT Category I / II at 11 kV at ₹7.65/kWh. Energy charge; ₹475/kVA/month demand charge excluded. APERC Retail Supply Tariff Order FY2025-26, continued for FY2026-27.
- Demand charges:** Excluded. Solar does not reduce contracted demand, so fixed and demand charges are unchanged by the plant.
- Tariff escalation:** 4% per year, applied to both the energy charge and the export rate.
- Generation:** 1,533 kWh per kWp in year one, degrading 0.5% per year.
- Unit split:** 90% self-consumed, 10% exported at ₹2.75/kWh. The plant is assumed to be sized to the site's daytime load.
- GST:** 5%, recovered in full as input tax credit. Figures are stated ex-GST. A buyer who cannot claim ITC should read the cost as ₹1.82 Cr.
- Depreciation:** 40% written-down value, Income Tax Act Appendix I. No additional depreciation under s.32(1)(iia), which s.115BAA forfeits.
- Tax:** 25.168% — s.115BAA at 22% plus 10% surcharge and 4% cess. Applied to savings and to depreciation alike.
- Loss set-off:** Early-year depreciation exceeds savings. The resulting loss is assumed to shelter the buyer's other business income in the same year.
- Commissioning:** Full first-year depreciation assumed, i.e. the plant is put to use on or before 30 September. Later commissioning halves the year-one charge, with the balance carried forward.
- Maintenance:** ₹600 per kWp per year from year 2, held flat. Year 1 is covered by the build.
- Inverter replacement:** Not modelled. A mid-life inverter replacement would reduce the figures shown.
- Financing:** None. All returns are unlevered — the return on the money invested, before any loan.
- Horizon:** 30 years, which runs beyond the module performance warranty.
- Discount rate:** 10% for net present value, discounted payback and levelised cost.
- Emissions:** Grid emission factor 0.82 kg CO₂ per kWh; tree equivalence at 45 trees per tonne is illustrative, not measured sequestration.

BASIS AND LIMITATIONS

This is an indicative financial analysis for a standard 450 kWp commercial rooftop plant. **It is not a quotation, an offer, or tax advice.** Pricing is subject to a site survey, roof condition, structure height, cable routing, evacuation distance and the APSPDCL sanction actually granted.

Tariffs, the depreciation block and the corporate tax rate are as we understand them at the date printed below and are subject to change by APERC and by the Finance Act. The tax treatment shown assumes a GST-registered company that has opted into s.115BAA and has sufficient other business income to absorb the early-year depreciation. **Have your chartered accountant confirm the tax position before you rely on it.**

Generation depends on site irradiance, shading, orientation, soiling and grid availability. Actual results will differ from this projection.